

Lecture 3: Wavefunction Methods for Electron Correlation

1. The Target

We want the last 1%:

$$E_{\text{corr}} = E_{\text{exact}} - E_{\text{HF}} < 0, \quad (1)$$

where “exact” means exact within the chosen one-particle basis. Unlike DFT, the methods in this lecture are *systematically improvable*: each defines a hierarchy that converges to the exact answer, and we can trade cost for accuracy in a controlled way. The price is steep scaling.

Before we start it is worth naming the two physically distinct regimes we will keep running into:

Weak (dynamic) correlation	Strong (static) correlation
short range, “Coulomb hole” equilibrium geometries one dominant determinant perturbation theory works	long range, near-degeneracy bond breaking, open shells many determinants of equal weight perturbation theory fails

There is no sharp boundary between them, but essentially every method below is built for one and fails at the other.

2. Second Quantization

Real-space Slater determinants are unwieldy. Since HF hands us an orthonormal orbital set, work in the occupation-number basis $|n_1 n_2 \cdots n_M\rangle$, $n_p \in \{0, 1\}$, with the HF determinant

$$|\Phi_{\text{HF}}\rangle = | \underbrace{1 \cdots 1}_{N \text{ occupied}} \underbrace{0 \cdots 0}_{\text{virtual}} \rangle. \quad (2)$$

Introduce fermionic ladder operators obeying

$$\{a_p, a_q^\dagger\} = \delta_{pq}, \quad \{a_p, a_q\} = \{a_p^\dagger, a_q^\dagger\} = 0. \quad (3)$$

Antisymmetry is now an operator identity rather than a bookkeeping rule, and $(a_p^\dagger)^2 = 0$ is Pauli. Convention: i, j, k, l occupied, a, b, c, d virtual, p, q, r, s either. The Hamiltonian becomes

$$\hat{H} = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{4} \sum_{pqrs} \langle pq || rs \rangle a_p^\dagger a_q^\dagger a_s a_r, \quad (4)$$

independent of N : all the particle-number information now lives in the state. The Slater–Condon rules follow from normal ordering; for instance $\langle \Phi_{\text{HF}} | \hat{H} | \Phi_{\text{HF}} \rangle = \sum_i h_{ii} + \frac{1}{2} \sum_{ij} \langle ij || ij \rangle$, as in Lecture 1.

3. The Excitation Hierarchy and How Big the Problem Is

Every basis state can be classified by its excitation rank relative to $|\Phi_{\text{HF}}\rangle$:

$$\hat{T}_1 = \sum_{ia} t_i^a a_a^\dagger a_i, \quad \hat{T}_2 = \frac{1}{4} \sum_{ijab} t_{ij}^{ab} a_a^\dagger a_b^\dagger a_j a_i, \quad \hat{T}_3, \hat{T}_4, \dots \quad (5)$$

For $N_\uparrow, N_\downarrow$ electrons in M spatial orbitals the full space has

$$N_{\text{det}} = \binom{M}{N_\uparrow} \binom{M}{N_\downarrow} \quad (6)$$

determinants. With $N_\uparrow = N_\downarrow = N$ and $M = cN$, Stirling gives

$$\log N_{\text{det}} \approx 2N \left[c \log \frac{c}{c-1} + \log(c-1) \right] = \mathcal{O}(N), \quad (7)$$

i.e. exponential growth. Even a toy case makes the point: four electrons in four spatial orbitals has 36 determinants at $M_S = 0$, of which CISD captures 27, so a quarter of the space already lies beyond doubles. There are two ways out, and this lecture covers both. Truncate the excitation rank and stay polynomial: MP2, CI, coupled cluster. Or keep the full space and either *compress* it (DMRG) or *sample* it (AFQMC).

4. Møller–Plesset Perturbation Theory (optional)

The cheapest correction. Partition the Hamiltonian using the Fock operator,

$$\hat{H}^{(0)} = \hat{F} = \sum_{pq} \varepsilon_p \delta_{pq} a_p^\dagger a_q, \quad \hat{H}^{(1)} = \hat{H} - \hat{F}, \quad (8)$$

so that $\hat{F}|\Phi_{\text{HF}}\rangle = (\sum_i \varepsilon_i) |\Phi_{\text{HF}}\rangle$ and every determinant is an eigenstate of $\hat{H}^{(0)}$. Rayleigh–Schrödinger perturbation theory then gives

$$E^{(0)} = \sum_i \varepsilon_i, \quad E^{(1)} = \langle \Phi_{\text{HF}} | \hat{H} - \hat{F} | \Phi_{\text{HF}} \rangle = -\frac{1}{2} \sum_{ij} \langle ij || ij \rangle, \quad E^{(0)} + E^{(1)} = E_{\text{HF}}. \quad (9)$$

The first-order energy is exactly the double-counting correction; MP1 gives nothing new. At second order,

$$E^{(2)} = \sum_{n \neq 0} \frac{|\langle \Phi_{\text{HF}} | \hat{H}^{(1)} | \Phi_n \rangle|^2}{E_0^{(0)} - E_n^{(0)}}. \quad (10)$$

Since $\hat{H}^{(1)}$ is a two-body operator, only singles and doubles contribute. The denominators are differences of orbital energies, $E_0^{(0)} - E_{ij}^{ab} = \varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b \equiv -\Delta_{ij}^{ab}$, the numerators are F_{ia} and $\langle ij || ab \rangle$, and the singles vanish by Brillouin’s theorem for canonical HF orbitals. Therefore

$$E_{\text{MP2}} = E_{\text{HF}} - \frac{1}{4} \sum_{ijab} \frac{|\langle ij || ab \rangle|^2}{\Delta_{ij}^{ab}} \quad (11)$$

Two remarks. First, $E^{(2)}$ is manifestly negative, since correlation always lowers the energy. Second, MP2 is size consistent, which is not obvious but follows from the fact that the numerator vanishes for excitations spanning noninteracting fragments.

Cost. The energy expression is $\mathcal{O}(o^2v^2)$, but the AO→MO transformation $(\mu\nu|\lambda\sigma) \rightarrow (ia|jb)$, done one index at a time, costs $\mathcal{O}(oM^4)$. Density fitting factorizes the integrals as $(\mu\nu|\lambda\sigma) \approx \sum_P B_{\mu\nu}^P B_{\lambda\sigma}^P$; the fit is robust, since the leading error cancels, and RI-MP2 keeps the $\mathcal{O}(N^5)$ asymptote with a far smaller prefactor: ~ 5000 basis functions in a day or two. Keep this factorization in mind — AFQMC will need exactly the same one.

5. When MP2 Fails (optional)

- **Accuracy is only DFT-like.** On broad thermochemistry benchmarks MP2 gives WTMAE ≈ 3.6 kcal/mol versus 3.7 for B3LYP — a fifth-order method tying a third-order one. The productive responses were to fold MP2 correlation into a functional (the double hybrids of Lecture 2) or to scale the spin components (SCS-MP2; dropping same-spin entirely gives comparable accuracy at $\mathcal{O}(N^4)$).
- **Small gaps.** If $\Delta_{ij}^{ab} \rightarrow 0$, (11) diverges. Rare for molecules (Koopmans says a vanishing gap means free ionization), fatal for metals: MP n simply does not exist for a metal.
- **Bond breaking and spin contamination.** RHF-MP2 diverges at dissociation; UHF-MP2 gives the right limit but a kink at the Coulson–Fischer point, where the HF orbital Hessian is discontinuous, and inherits spin contamination ($\langle \hat{S}^2 \rangle = 0.38$ for benzene, a closed-shell molecule). Orbital-optimized MP2 removes both.

6. Configuration Interaction

The variational alternative. Write

$$|\Psi_{\text{CI}}\rangle = \sum_I c_I |\Phi_I\rangle, \quad E_{\text{CI}} = \min_{\mathbf{c}} \frac{\mathbf{c}^\dagger \mathbf{H} \mathbf{c}}{\mathbf{c}^\dagger \mathbf{c}} \iff \mathbf{H} \mathbf{c} = E \mathbf{c}, \quad (12)$$

with H_{IJ} from the Slater–Condon rules. Full CI (I = all determinants) is exact in the basis but limited to $\sim 10^9$ – 10^{10} determinants, meaning about 18 electrons in 18 orbitals. Use it whenever you can; it is the reference against which everything else is judged.

Truncating gives CIS, CISD, CISDT, The block structure of the CISD Hamiltonian, ordered as (reference, singles, doubles), is dictated by two facts: $\hat{H}$ is a two-body operator, and Brillouin’s theorem kills the reference–singles block:

$$\mathbf{H}_{\text{CISD}} = \begin{pmatrix} E_{\text{HF}} & \mathbf{0} & \langle ij||ab \rangle \\ \mathbf{0} & \mathbf{H}_{SS} & \mathbf{H}_{SD} \\ \langle ab||ij \rangle & \mathbf{H}_{DS} & \mathbf{H}_{DD} \end{pmatrix}. \quad (13)$$

Because the reference–singles block vanishes, singles improve the ground state only indirectly, through their coupling to doubles; and if we keep singles alone (CIS) the reference decouples entirely, which is why CIS is an excited-state method (Lecture 4), not a correlation method.

The two-state model. Nearly everything about correlation is visible in

$$\mathbf{H} = \begin{pmatrix} E_{\text{HF}} & V \\ V & E_D \end{pmatrix}, \quad \Delta_D = E_D - E_{\text{HF}} > 0. \quad (14)$$

Exactly, $E_{\pm} = \bar{E} \pm \sqrt{(\Delta_D/2)^2 + V^2}$, and for $|V| \ll \Delta_D$,

$$E_- \simeq E_{\text{HF}} - \frac{V^2}{\Delta_D}, \quad \frac{c_D}{c_0} \simeq -\frac{V}{\Delta_D}. \quad (15)$$

The correlation energy is second order in the coupling and always negative; the weight of the excited configuration is first order. When $|V| \sim \Delta_D$ the expansion fails and we are in the strongly correlated regime, which is exactly what happens to $|\sigma_g^2\rangle$ and $|\sigma_u^2\rangle$ as H_2 is stretched.

7. Size Consistency: Why Truncated CI Died

Consider two noninteracting fragments A and B , each described at the doubles level, $|\Psi_X\rangle = (1 + \hat{T}_2^X)|\Phi_X\rangle$. The correct supersystem wavefunction is the product

$$|\Psi_A\rangle|\Psi_B\rangle = \left(1 + \hat{T}_2^A + \hat{T}_2^B + \hat{T}_2^A\hat{T}_2^B\right)|\Phi_A\Phi_B\rangle, \quad (16)$$

which contains a *quadruple* excitation, a simultaneous double on each fragment. CID on the supersystem, $(1 + \hat{T}_2^{AB})|\Phi_{AB}\rangle$, cannot represent it, so

$$E_{\text{CID}}(A+B) \neq E_{\text{CID}}(A) + E_{\text{CID}}(B). \quad (17)$$

The error grows with system size: for n noninteracting H_2O molecules, the CISD correlation energy per molecule decays instead of staying constant. This single flaw is why truncated CI is essentially never used for ground states today.

8. Coupled Cluster Theory

The fix is to make the disconnected products appear automatically, by exponentiating:

$$|\Psi_{\text{CC}}\rangle = e^{\hat{T}}|\Phi_{\text{HF}}\rangle, \quad \hat{T} = \hat{T}_1 + \hat{T}_2 + \dots \quad (18)$$

For noninteracting fragments $\hat{T} = \hat{T}_A + \hat{T}_B$ with $[\hat{T}_A, \hat{T}_B] = 0$, so $e^{\hat{T}} = e^{\hat{T}_A}e^{\hat{T}_B}$ and the wavefunction factorizes exactly. CCD is size consistent at *any* truncation level; CID is not.

We cannot use the variational energy, because $\langle\Phi|e^{\hat{T}^\dagger}\hat{H}e^{\hat{T}}|\Phi\rangle$ does not truncate and costs $\mathcal{O}(e^N)$, so we define the *projected* energy and amplitude equations, using intermediate normalization, $\langle\Phi_{\text{HF}}|\Psi_{\text{CC}}\rangle = 1$:

$$E_{\text{CC}} = \langle\Phi_{\text{HF}}|\hat{H}e^{\hat{T}}|\Phi_{\text{HF}}\rangle, \quad \langle\Phi_\mu|(\hat{H} - E)e^{\hat{T}}|\Phi_{\text{HF}}\rangle = 0 \quad \forall \mu \in \{\text{retained excitations}\}. \quad (19)$$

The number of equations equals the number of amplitudes. The energy is not variational and not an upper bound. That is the price of size consistency and polynomial cost. Because $\hat{H}$ is two-body, the expansion terminates:

$$E_{\text{CCSD}} = E_{\text{HF}} + \frac{1}{4} \sum_{ijab} \langle ij || ab \rangle \left(t_{ij}^{ab} + 2t_i^a t_j^b \right). \quad (20)$$

Cluster analysis: how CC relates to CI. Expanding $e^{\hat{T}_1+\hat{T}_2}$ and matching coefficients with the CI expansion $|\Psi\rangle = (1 + \hat{C}_1 + \hat{C}_2 + \dots)|\Phi_{\text{HF}}\rangle$ gives

$$c_i^a = t_i^a, \quad c_{ij}^{ab} = t_{ij}^{ab} + t_i^a t_j^b - t_i^b t_j^a. \quad (21)$$

The second equation is the whole story: the CI double coefficient is a *connected* part t_{ij}^{ab} (genuine two-electron correlation) plus a *disconnected* antisymmetrized product of two independent singles. Since disconnected products dominate at large system size, the t amplitudes are far more compact than the c amplitudes, and the CC hierarchy converges much faster.

The hierarchy. CCS is just HF (Thouless: $e^{\hat{T}_1}$ acting on a determinant is another determinant). CCSD is exact for two electrons, correct through MP3, and costs $\mathcal{O}(o^2 v^4)$, the bottleneck being the particle–particle ladder $\sum_{cd} \langle ab || cd \rangle t_{ij}^{cd}$. Adding the leading fourth-order triples correction non-iteratively gives **CCSD(T)**, the “gold standard”:

$$\mathcal{O}(o^3 v^4), \quad \sim 1 \text{ kcal/mol for main-group thermochemistry.} \quad (22)$$

Property	HF	DFT	MP2	CCSD(T)
IPs and EAs	± 0.5 eV	± 0.2 eV	± 0.2 eV	± 0.05 eV
Bond lengths	-1%	± 1 pm	± 1 pm	± 0.5 pm
Vibrational frequencies	$+10\%$	$+3\%$	$+3\%$	± 5 cm^{-1}
Barrier heights	$+30\text{--}50\%$	-25%	$+10\%$	± 2 kcal/mol
Bond energies	-50%	± 3 kcal/mol	± 10 kcal/mol	± 1 kcal/mol

Practical limits: ~ 100 electrons and 500 basis functions for CCSD(T) in about a week. Local approximations (DLPNO-CCSD(T)) push this to hundreds of atoms, at the cost of a less controlled error.

9. Where Coupled Cluster Breaks

- **More than one strongly correlated pair.** CCSD is exact for one electron pair; N_2 dissociation has three, and RCCSD fails badly (non-variationally, going *below* the exact energy). Interestingly the T amplitudes are fine; it is the projected energy functional that misbehaves. UCCSD gets the limit right but errs by ~ 6 kcal/mol in the spin-recoupling region. And (T) makes strong correlation *worse*, since it inherits the divergent MP4-like denominators.
- **The T_1 diagnostic is not a correlation diagnostic.** Large T_1 usually signals bad orbitals (an orbital rotation), not a multireference system, and better orbitals often fix it: CCSD(T) harmonic frequencies across 95 molecules go from 48 to 8 cm^{-1} RMSD for open shells on switching UHF orbitals for κ -OOMP2 ones.
- **Transition metals.** ~ 5 kcal/mol errors on 3d bond dissociation energies; higher excitations matter for both dynamic and strong correlation.

10. When One Reference Is Not Enough

If several determinants have comparable weight, expand around several determinants. The standard route:

1. Choose an *active space* of chemically relevant orbitals and electrons, CAS(ne,mo); treat it exactly.
2. Optimize orbitals *and* CI coefficients simultaneously,

$$E_{\text{CASSCF}} = \min_{\mathbf{C}} \min_{\{\text{CI}\}} \frac{\langle \Psi_{\text{CI}} | \hat{H} | \Psi_{\text{CI}} \rangle}{\langle \Psi_{\text{CI}} | \Psi_{\text{CI}} \rangle}. \quad (23)$$

3. Add the missing dynamic correlation on top: CASPT2, NEVPT2, MRCI, MRCC.

CASSCF is size consistent but exponential in the active space, the optimization is riddled with local minima, and the choice of active space is an art rather than an algorithm, so these calculations are not reproducible in the way a CCSD(T) number is. Selected CI (CIPSI and its modern descendants) attacks the same problem differently: start from a small determinant set, score each external determinant by the perturbative estimate $\delta E_J^{(2)} = |\langle \Phi_J | \hat{H} | \Psi_{\text{var}} \rangle|^2 / (E_{\text{var}} - \langle \Phi_J | \hat{H} | \Phi_J \rangle)$, add the important ones, and iterate. It reaches FCI quality for small systems and is the state of the art in brute force, but it is still exponential. The next two sections take the opposite tack: instead of choosing which determinants to keep, change the representation.

11. Compressing the Wavefunction: DMRG

Write the full CI vector as a tensor with one index per orbital,

$$|\Psi\rangle = \sum_{n_1 \cdots n_M} c_{n_1 \cdots n_M} |n_1 \cdots n_M\rangle, \quad n_p \in \{\text{empty}, \uparrow, \downarrow, \uparrow\downarrow\}, \quad (24)$$

so c has 4^M entries. DMRG replaces it by a *matrix product state*: one matrix per orbital, contracted along a chain,

$$c_{n_1 \cdots n_M} = \sum_{a_1 \cdots a_{M-1}} A_{a_1}^{n_1} A_{a_1 a_2}^{n_2} \cdots A_{a_{M-1}}^{n_M}, \quad \dim(a_i) \leq D. \quad (25)$$

The parameter count falls from 4^M to $\mathcal{O}(MD^2)$, and the *bond dimension* D is a single systematically improvable knob: $D = 1$ is a product state, D exponentially large is exact FCI. The A 's are optimized variationally by sweeping along the chain, solving a local eigenvalue problem at each orbital with the rest fixed, at $\mathcal{O}(M^3 D^3)$ cost.

Why does small D ever suffice? Because D is controlled by *entanglement* rather than by excitation rank. Cut the chain between orbitals p and $p+1$ and take the von Neumann entropy $S = -\text{Tr } \rho \log \rho$ of the reduced density matrix; an MPS with bond dimension D can carry at most $S = \log D$. Systems whose entanglement stays bounded across every cut — one-dimensional chains, and by extension the strongly correlated orbitals of a polyene or an Fe-S cluster once they are localized and ordered sensibly along a line — need only modest D . That is how DMRG reaches active spaces of 50–100 orbitals where CASSCF stops near 18.

The failure mode is geometric: entanglement across a cut through a two- or three-dimensional system grows with the *area* of the cut, so D grows exponentially with width and orbital ordering along the artificial chain becomes critical. DMRG is also a static-correlation tool — it wants a few strongly entangled orbitals, not the thousands of weakly correlated virtuals where dynamic correlation lives — so it inherits CASSCF’s division of labor, with perturbation theory layered on top (DMRG-NEVPT2, DMRG-CASPT2).

12. Sampling the Wavefunction: AFQMC

The other option is to stop storing the wavefunction at all. Any trial state with nonzero overlap is projected onto the ground state by imaginary time,

$$|\Psi_0\rangle \propto \lim_{\tau \rightarrow \infty} e^{-\tau \hat{H}} |\Psi_T\rangle, \quad \langle \Psi_T | \Psi_0 \rangle \neq 0, \quad (26)$$

so everything reduces to applying $e^{-\Delta\tau \hat{H}}$ many times. The obstruction is that $\hat{H}$ is two-body, and $e^{-\Delta\tau \hat{V}}$ does not map a determinant to a determinant. Write the two-body part as a sum of squares of one-body operators — which the Cholesky or density-fitting factorization of the ERIs already hands us, $\hat{V} = \frac{1}{2} \sum_{\gamma} \hat{v}_{\gamma}^2$ — and linearize each square with a Gaussian integral, the Hubbard–Stratonovich transformation:

$$e^{-\Delta\tau \hat{v}^2/2} = \int dx \frac{e^{-x^2/2}}{\sqrt{2\pi}} e^{\sqrt{-\Delta\tau} x \hat{v}}. \quad (27)$$

The two-body operator is gone, traded for an integral over *auxiliary fields* x that we do by Monte Carlo. Every term in the integrand is a one-body exponential, and by Thouless’ theorem that maps a determinant to a determinant. The algorithm is thus a random walk over *non-orthogonal* determinants $|\phi_k\rangle$ with weights w_k , and the energy comes from the mixed estimator

$$E = \frac{\sum_k w_k \langle \Psi_T | \hat{H} | \phi_k \rangle / \langle \Psi_T | \phi_k \rangle}{\sum_k w_k}. \quad (28)$$

The cost is $\mathcal{O}(M^3)$ – $\mathcal{O}(M^4)$ per sample with statistical error falling as $N_s^{-1/2}$, and the walkers are independent, so the method is embarrassingly parallel — the cheapest asymptotic scaling of anything in this lecture that reaches CCSD(T) accuracy.

The phase problem. For a general electronic Hamiltonian the one-body propagators are complex, the overlaps $\langle \Psi_T | \phi_k \rangle$ wander over the complex plane, and the signal-to-noise ratio decays exponentially in τ : this is the fermionic sign problem in AFQMC guise. The standard cure is the *phaseless* approximation, constraining each walker’s overlap phase toward the trial state. That restores polynomial cost at a price worth stating plainly: ph-AFQMC is no longer variational, and its residual error depends on $|\Psi_T\rangle$. In practice a single HF or DFT determinant already delivers roughly CCSD(T)-quality main group thermochemistry, and the error falls systematically as the trial improves — a CASSCF, selected-CI, or DMRG $|\Psi_T\rangle$ carries ph-AFQMC into the strongly correlated regime. So the last two sections are complements, not rivals: DMRG supplies the compact strongly correlated trial, AFQMC the dynamic correlation in a large basis.

13. Summary

- Second quantization makes antisymmetry automatic and the Hamiltonian particle-number independent.
- MP2 is the cheapest useful correlation method: $\mathcal{O}(N^5)$, size consistent, but only DFT-accurate and unusable for small gaps.
- Truncated CI is variational but not size consistent, and is therefore dead for ground states.
- Coupled cluster fixes size consistency by exponentiation; CCSD(T) is the gold standard at $\mathcal{O}(N^7)$.
- All of the above truncate the excitation rank and assume one dominant determinant. The alternative is to keep the full space in a different representation: DMRG compresses it, and is variational, deterministic, and limited by entanglement; AFQMC samples it, and is non-variational, stochastic, and limited by the phase problem. Used together they cover both correlation regimes.
- Strong correlation is nonetheless still the open problem, which is where Lecture 5 will pick up.